FoN – Glia Forms and Function

Lecture
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Learning Outcomes

By the end of this lecture, you should be able to:

- Identify the major types of glial cells
- Compare myelinating cells in the CNS and PNS
- Explain key functions of astrocytes
- Describe the role of microglia in CNS defence
- Explain glial responses to nervous system injury

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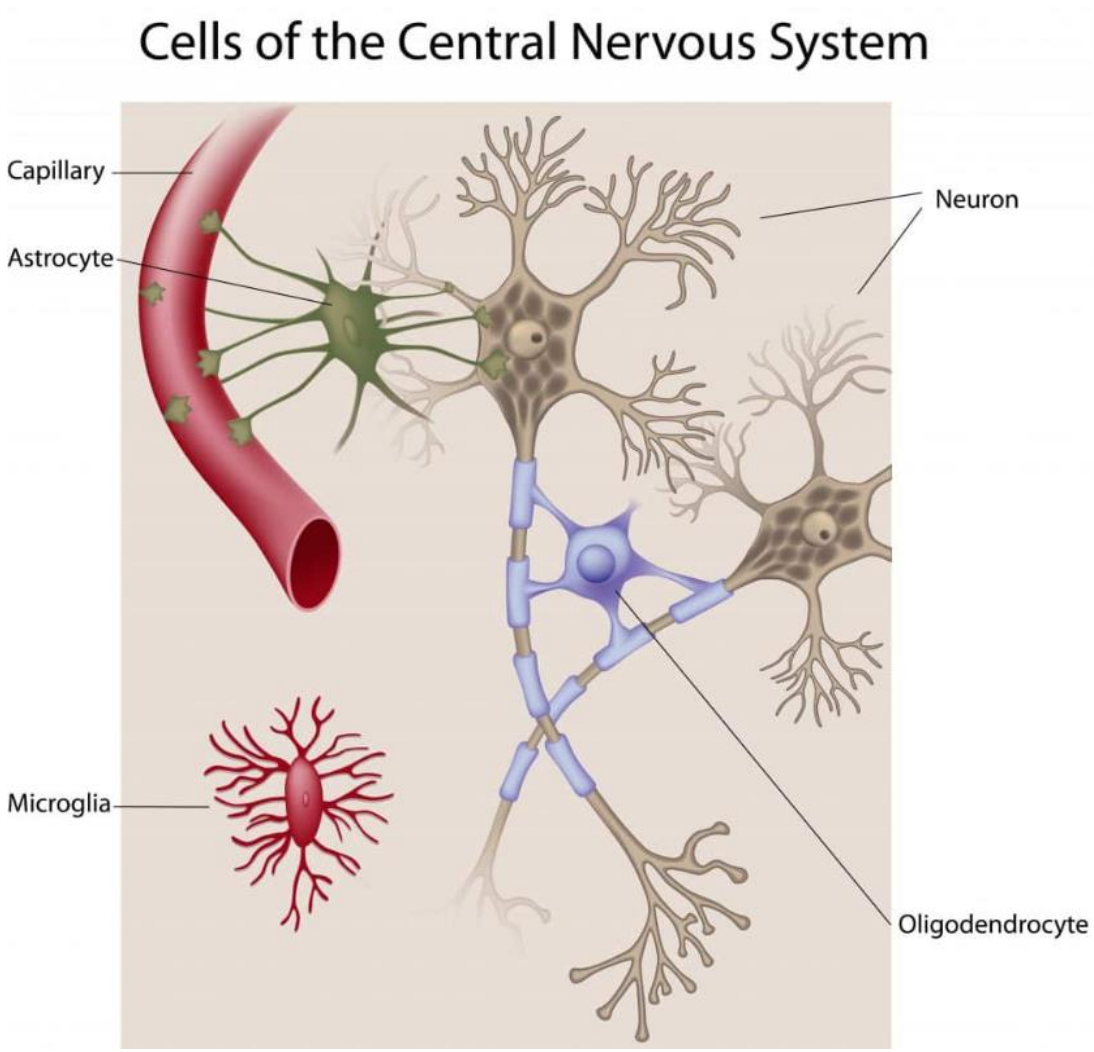
Neurons do not work alone

The nervous system contains neurons AND Glia

Glia provide:

- Physical support
- Metabolic support
- Electrical insulation
- Guidance during development
- Regulation of neuronal signalling
- Defence and response to injury

Neuron function depends on its cellular environment.



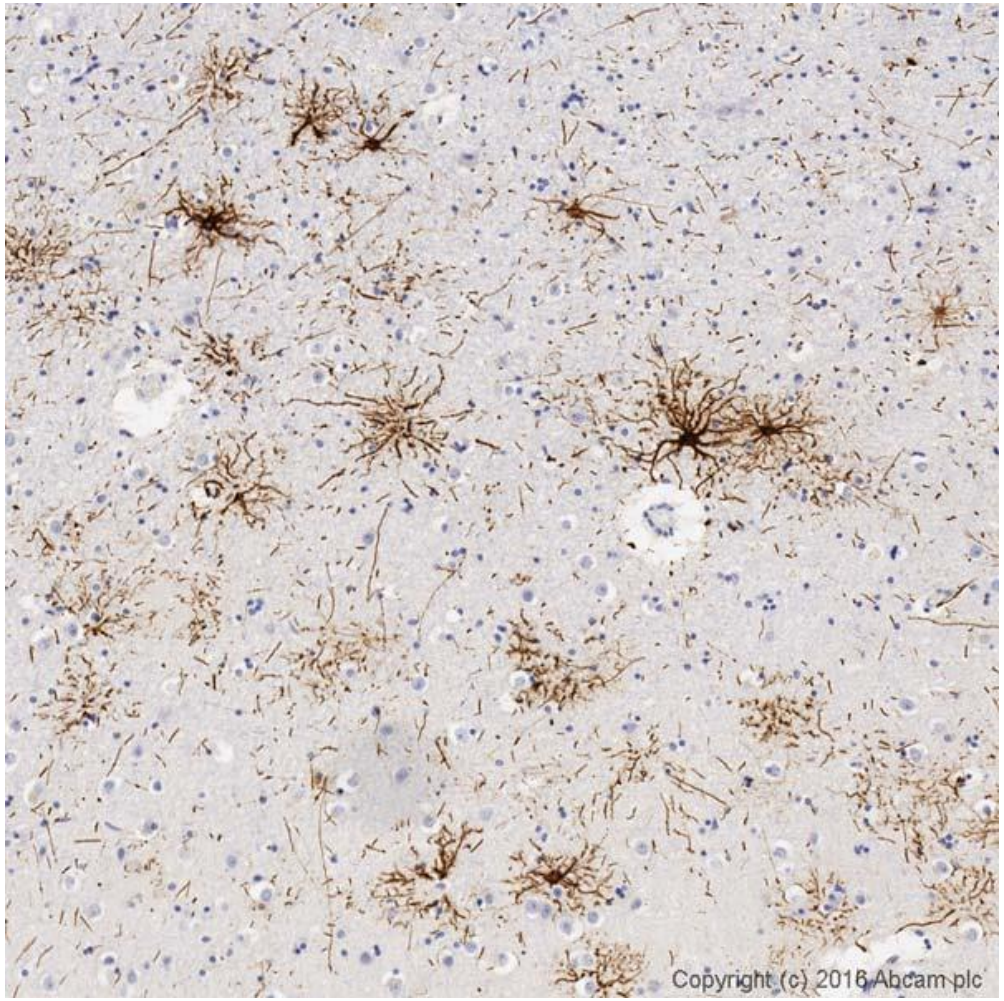
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What are glial cells?

Glia = ‘glue’

Non-excitable cells
NOT ‘seen’ by electrophysiological techniques

How can we identify different glial cells?
1. Morphology
2. Cell-specific markers
→ Immunohistochemistry (IHC)



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Major Glial Cell Types

- Macroglia
  - Schwann cells (in periphery)
  - Oligodendrocytes
  - Ependymal cells
  - Astrocytes

  } myelin-forming cells

- Microglia

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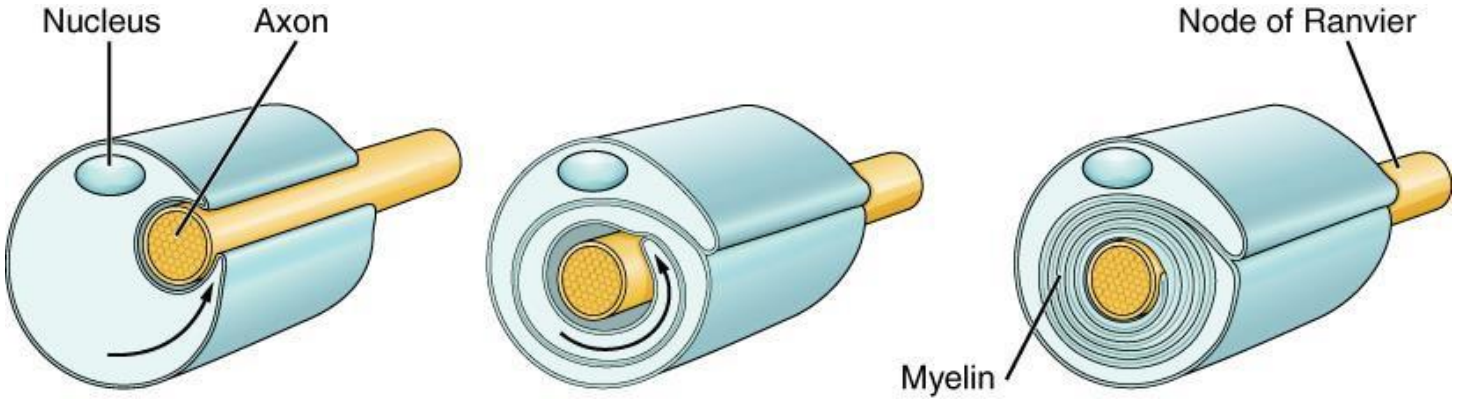
Section 2 – MYELIN-FORMING CELLS

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Schwann cells and oligodendrocytes

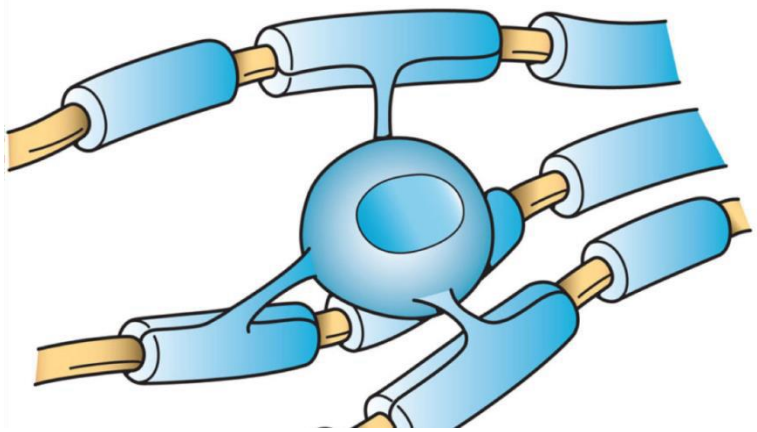
Schwann cell

- **PNS**
- One Schwann cell myelinates a single axon segment



Oligodendrocyte

- **CNS**
- One oligodendrocyte can myelinate **multiple axons — as many as ~30**



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Wikimedia Commons CC BY 3.0 Oligodendrocyte illustration Artwork by Holly Fischer

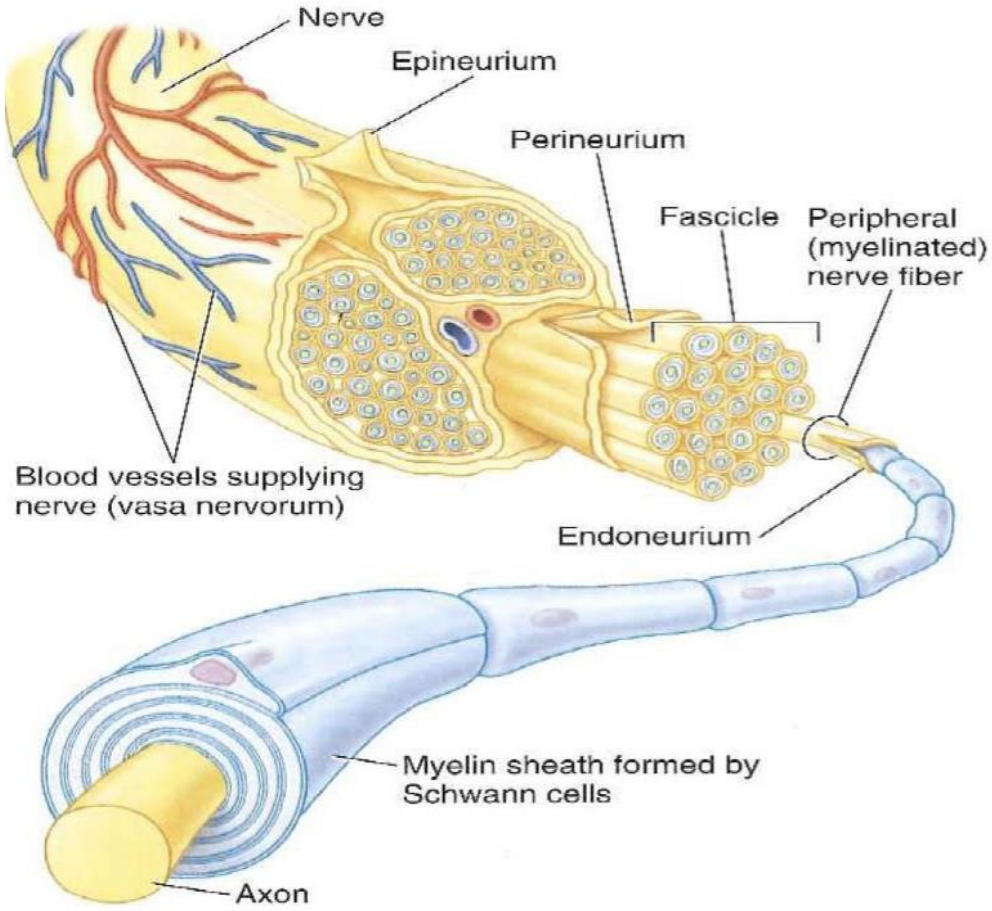
Schwann cell: formation of myelin

How does a Schwann cell form myelin?

Formation of the myelin sheath

Schwann cell membrane wraps concentrically around the axon

1. Schwann cell contacts axon
2. Membrane begins to wrap
3. Repeated membrane layers accumulate
4. A myelin sheath is formed



contact → wrapping → multiple membrane layers → myelin sheath.

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Myelinating cells and axon regeneration

- **Role in axon regeneration**

- **Peripheral nervous system: Regeneration possible**

Schwann cells + endoneurial tube

→ encourage axonal growth

- **Central nervous system: Regeneration very limited**

•No endoneurial tube

•Growth actively prevented by signalling in the CNS

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Section 3 – EPENDYMAL CELLS

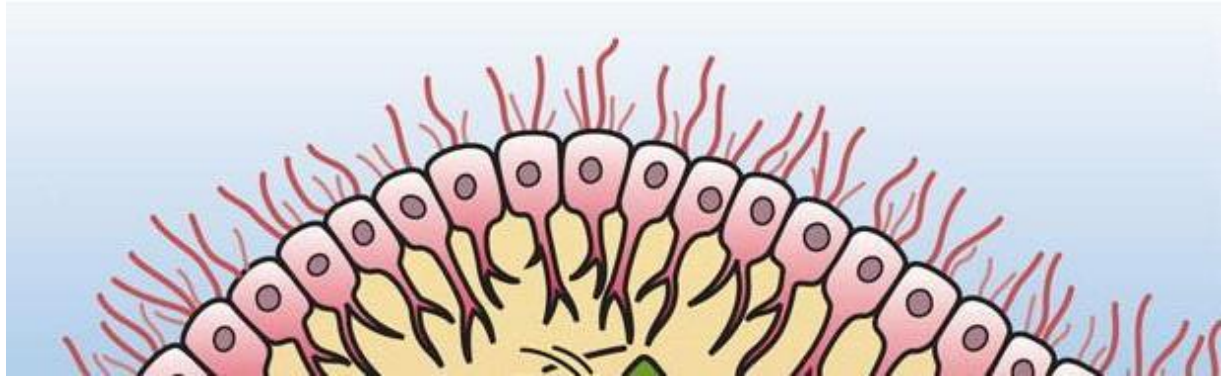
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Ependymal cells

Location

•Walls of brain ventricles

•Central canal of spinal cord



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Functions

•Involved in cerebrospinal fluid (**CSF**) production

•Cilia help circulate CSF

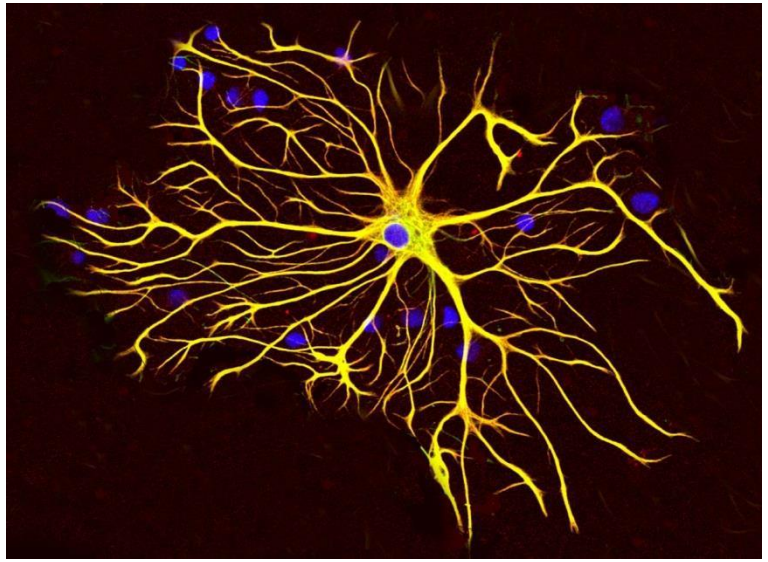
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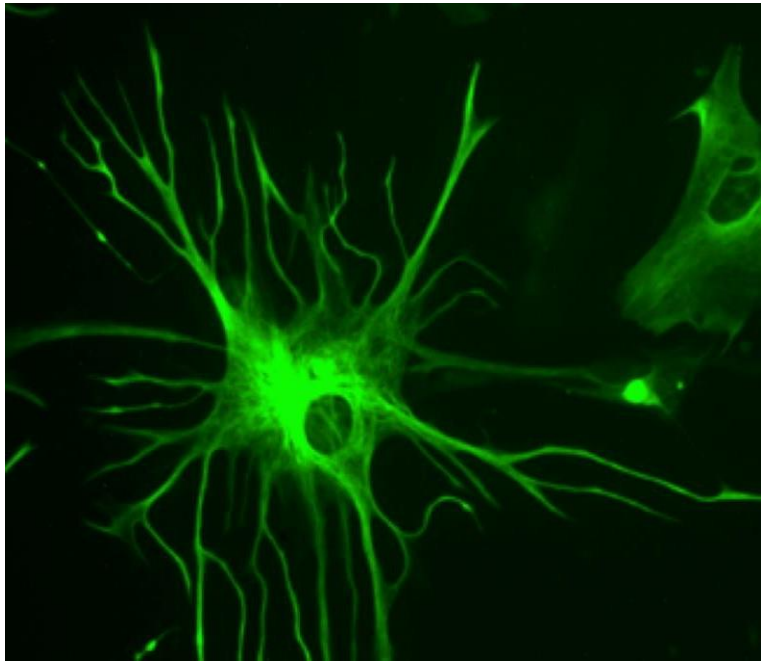
Section 4 – ASTROCYTES

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Astrocytes: form

- Large and numerous glial cells
- **Stellate morphology**
- Long cellular processes
- Processes may look like dendrites
- **Marker: GFAP,** Glial fibrillary acidic protein

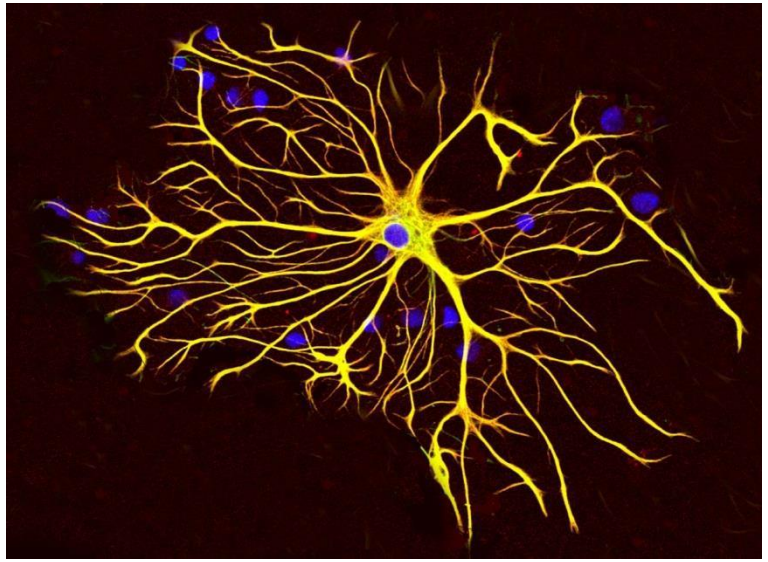


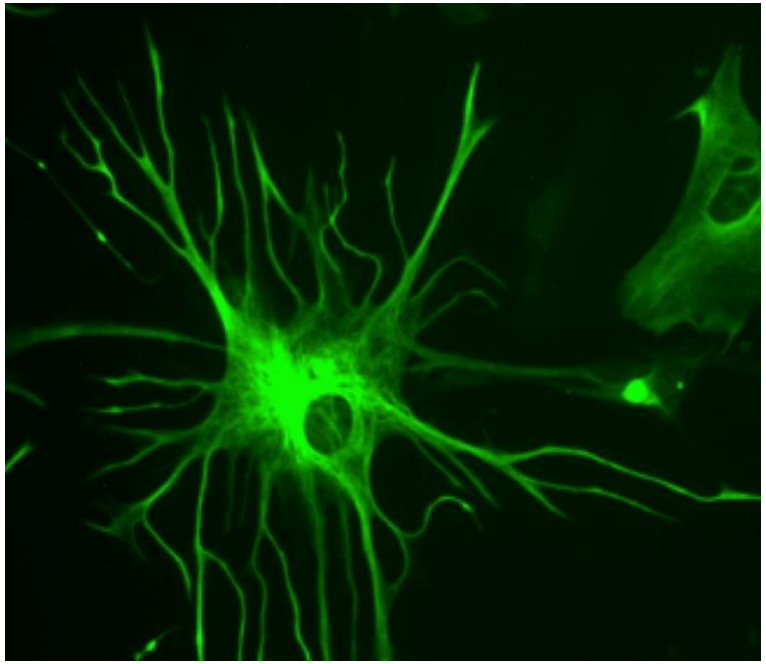


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Astrocytes: general functions

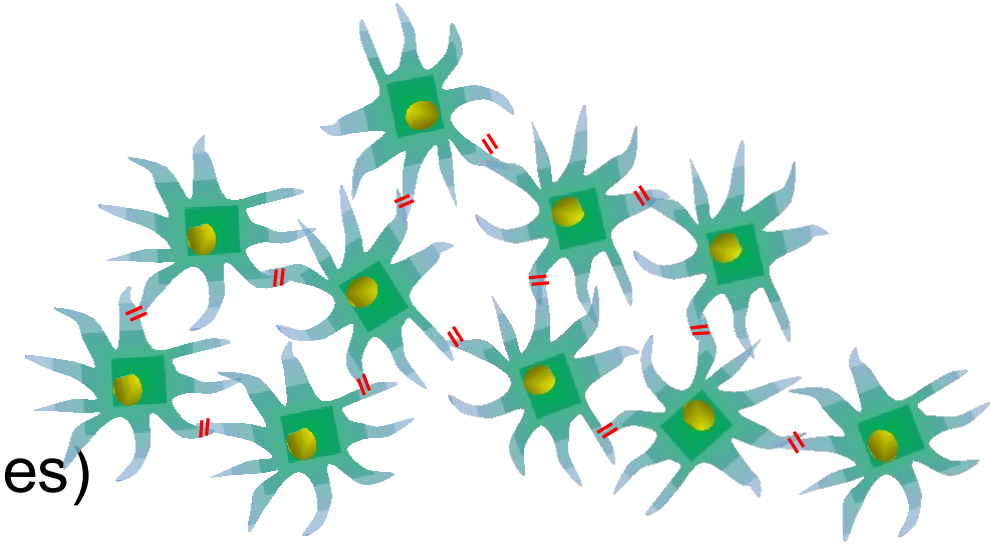
- **Neuronal guidance during development**
- Contribute to the architecture of the brain and spinal cord
- Astroglial stem cells: adult neurogenesis & gliogenesis



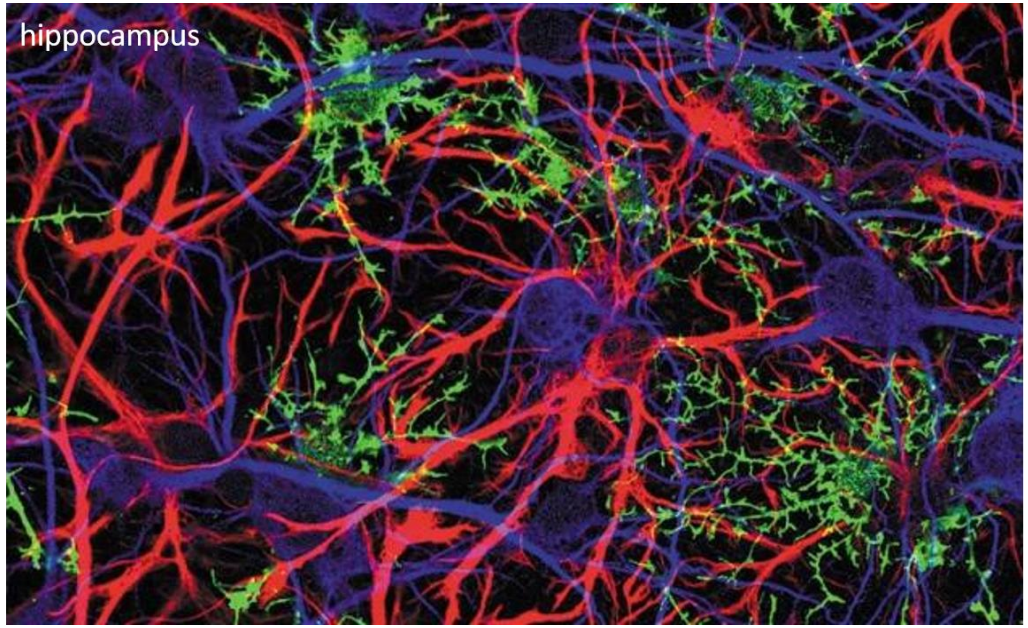


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Astrocytic syncytium



- Astrocytes form **large networks**
- Connected through **gap junctions** (electrical synapses)
- physically and electrically connected (**unlike neurons**)

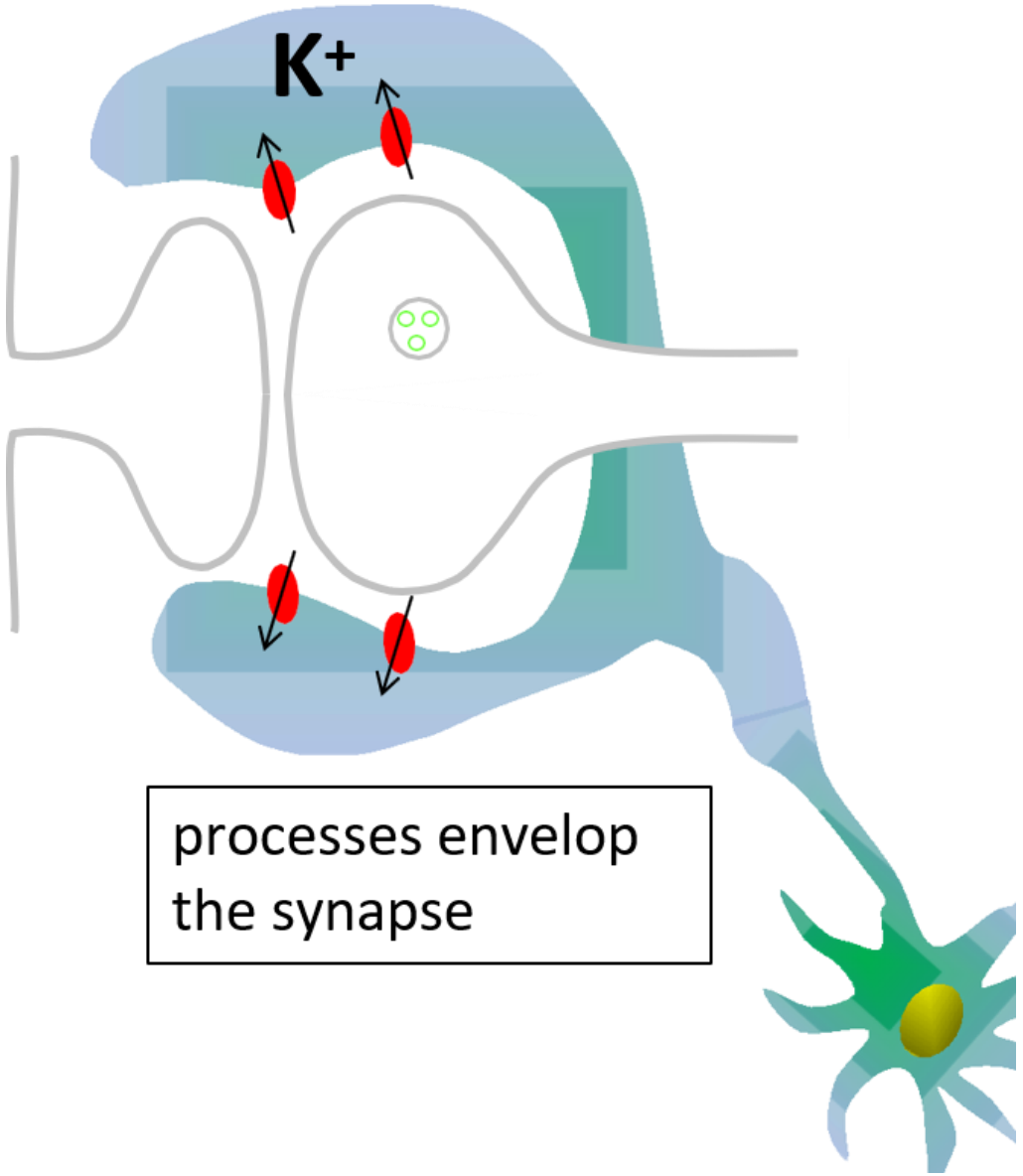


Astrocytes (red)
Neurons (blue)

One astrocyte may contact ~100,000 neuronal synapses

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Astrocytes: homeostatic mechanisms



Neuronal firing

↓

K^+ increases in extracellular fluid (ECF)

↓

Astrocytes take up K^+

↓

Prevent excessive K^+ accumulation

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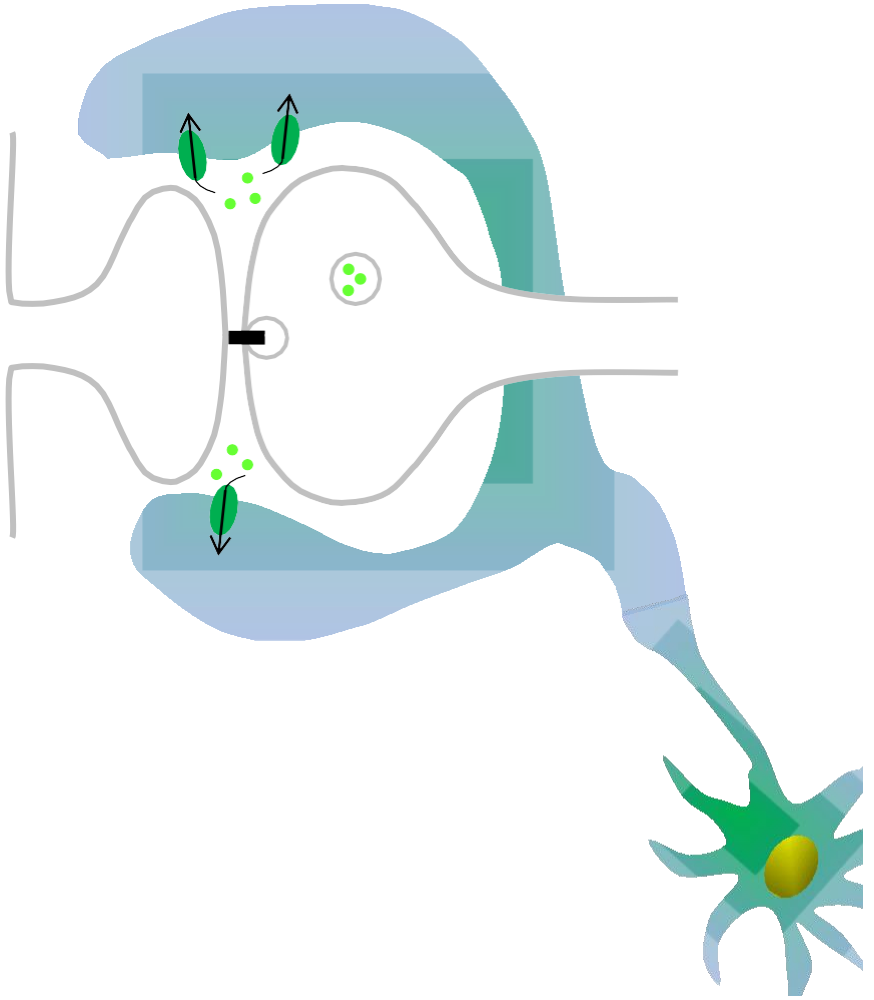
Astrocytes: controlling the synapse

- Astrocyte processes **envelop the synapse**
- Astrocytes express:

  Neurotransmitter transporters

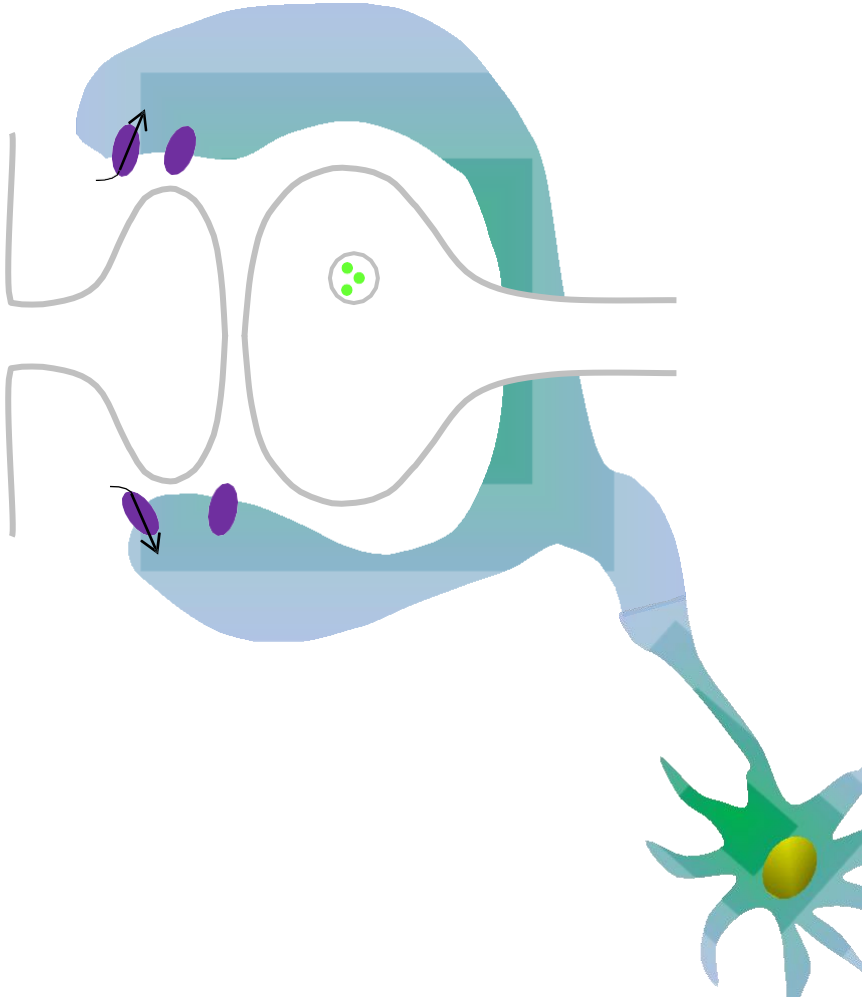
  ↓

  Prevent neurotransmitter diffusion outside the synapse



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Astrocytes: protecting the synapse



Astrocytes express **various transporters**

↓

Help prevent entry from outside the synapse

↓

Protect the synapse from the surrounding ECF

↓

Maintain the **synaptic microenvironment**

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Astrocytes: homeostatic functions

K^+ buffering

Prevents K^+ accumulation following neuronal firing

Controlling the synapse

Limits neurotransmitter diffusion

Protecting the synapse

Maintains the synaptic microenvironment

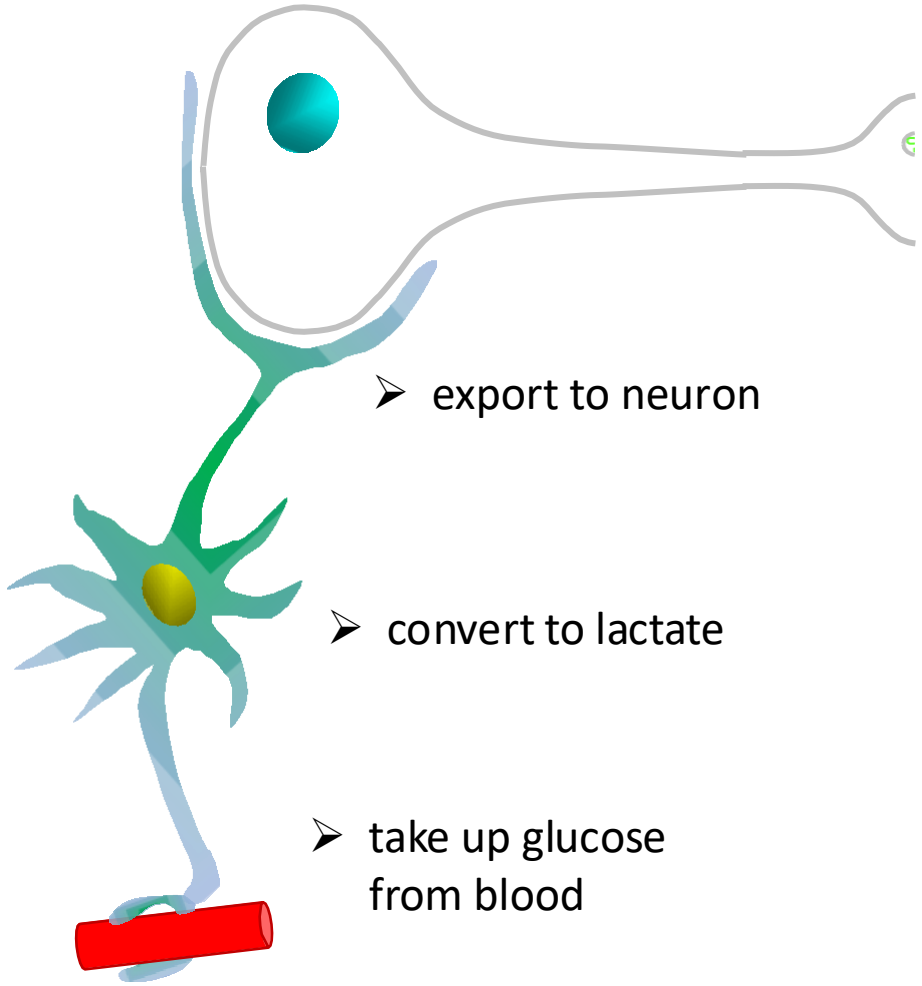
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Astrocytes: metabolic support

- Astrocytic **“foot” processes** envelop: Capillaries & Neurons
- **Astrocytes provide energy to neurons**
- **Metabolic support:**

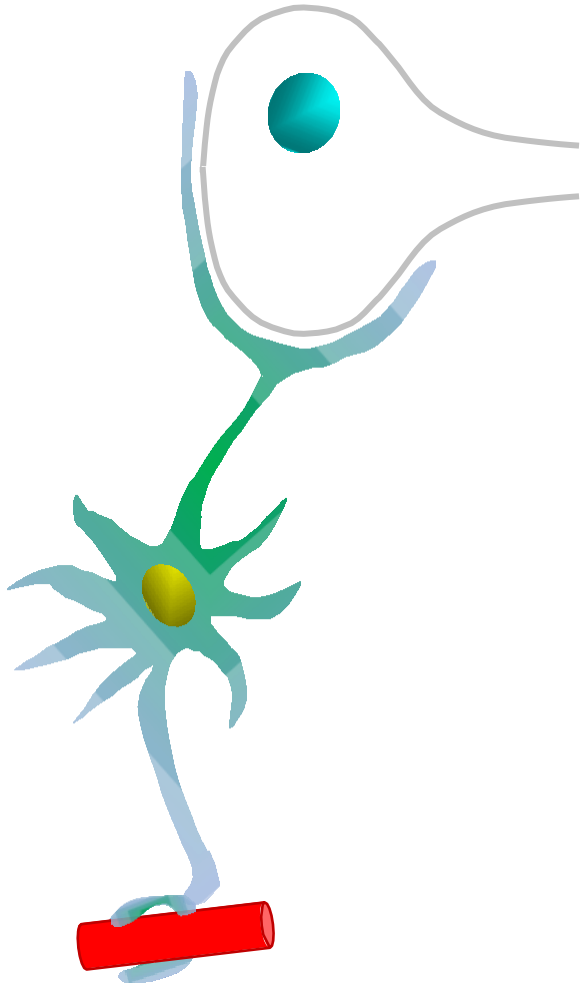
Blood glucose
↓
Astrocyte
↓
converted to **lactate**
↓
exported to neuron

- **Important for highly active neurons**



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Astrocytes: neurovascular coupling



- Astrocytes may modulate: **Cerebral blood flow**
- Changes in blood flow can provide :**Indirect measure of neuronal activity**
- **Application:**

  fMRI: functional magnetic resonance imaging

  BOLD contrast (Blood-Oxygen-Level Dependent)

→ measures the haemodynamic response

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Astrocytes: responding to injury / insult

❖ Damaged cells release: **ATP + glutamate**

- **Hypertrophy:** increase in size
- **Proliferation:** increase in cell number (**Reactive astrogliosis)**
- **GFAP expression ↑**
- **Glial scar formation around injured CNS**

❖ Astrocytes can produce: **Inhibitory molecules**

↓

Inhibit growth of damaged or severed axons

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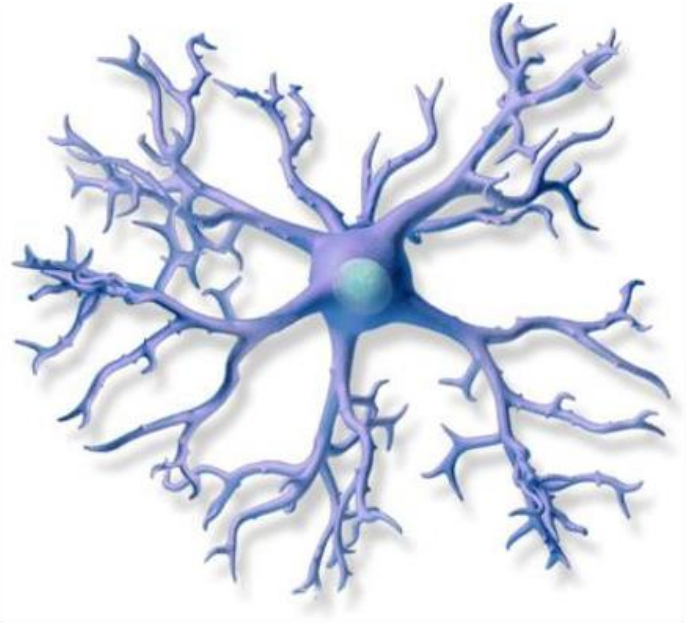
Section 5 – MICROGLIA

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Microglia: form

- Smallest glial cell type
- ~10% of total glial cells
- **Resident immune-competent cells in the CNS**
- arise from bone marrow monocytes

$\Rightarrow$ migrate to nervous system during development



modified from
Blausen.com staff. "Blausen gallery 2014".
Wikiversity Journal of Medicine.
DOI:10.15347/wjm/2014.010. ISSN 20018762.

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Microglia: responding to danger

In adults:

- Motile processes monitor the environment
- Danger signals (disease) or injury → trigger transformation

RESTING / SURVEILLANT →ACTIVATED →PHAGOCYTIC

Functions

•CNS immune defence

•Injury defence and repair

•Reactive microgliosis



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Microglial activation

- Microglia express: Immunological & Neurotransmitter receptors
- **Activation stimuli:**

1. Tissue injury
2. Pathogen-derived triggers (Lipopolysaccharide (LPS), Viral coat proteins)
3. Neuronal signalling (**ATP**)

- **Response is graded**
- **Microglial activation result in:**

1. Proliferation
2. Nitric oxide synthase (NOS) ↑
3. Release of pro-inflammatory cytokines: TNFα; IL-1β

↓

Neuroinflammatory response

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Neuroinflammation and CNS disorders

- Activated microglia can be **neurotoxic**
- Long-lasting microglial activation is linked to **pathological processes**, possible involved in Alzheimer's disease, Parkinson's disease and Ischaemic brain damage
- Activated microglia

↓

Astrocyte activation

↓

Disruption of normal homeostatic mechanisms(a definition of disease)

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Summary

Myelinating cells
PNS: Schwann cells
CNS: Oligodendrocytes
important differences for axon regeneration

Astrocytes
important roles in neurotransmission and neuronal function

Microglia
→ CNS immune cells
→ transform in response to danger signals

Reactive gliosis
CNS injury involves changes in **astrocytes + microglia**

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MCQ 1

Which glial cell forms myelin in the peripheral nervous system?

A. Astrocyte
B. Microglia
C. Schwann cell
D. Ependymal cell

MCQ 2

What is a major role of the Schwann cell and endoneurial tube after peripheral nerve injury?

A. They prevent axonal growth.
B. They encourage and guide axonal growth.
C. They produce cerebrospinal fluid.
D. They release pro-inflammatory cytokines.

MCQ 3

Where are ependymal cells found?

A. Around peripheral nerves only
B. In the walls of brain ventricles and the central canal of the spinal cord
C. Only in skeletal muscle
D. Only around synapses

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MCQ 4

What does GFAP stand for?

A. Glial fibrillary acidic protein
B. General fibre-associated protein
C. Glial functional axonal protein
D. General fibrillary adhesion protein

MCQ 5

Astrocytes are connected to each other through:

A. Myelin sheaths
B. Endoneurial tubes
C. Gap junctions
D. Cilia

MCQ 6

What happens to extracellular K^+ following neuronal firing?

A. It may increase.
B. It always decreases to zero.
C. It is converted into lactate.
D. It becomes a neurotransmitter.

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MCQ 7

How do astrocytes help control extracellular K^+?

A. They release more K^+ into the ECF.
B. They take up K^+ ions.
C. They convert K^+ into glucose.
D. They prevent neurons from firing.

MCQ 8

What is one function of astrocyte neurotransmitter transporters?

A. To increase neurotransmitter diffusion outside the synapse
B. To prevent neurotransmitter diffusion outside the synapse
C. To produce CSF
D. To cause axon degeneration

MCQ 9

Neurovascular coupling in this lecture is associated with:

A. Cerebral blood flow
B. Peripheral nerve regeneration only
C. CSF circulation only
D. Schwann-cell proliferation only

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SAQ 1

Give three ways in which astrocytes help maintain the environment around a synapse. [3 marks]

1: taking up neurotransmitter via **transporters**
2: wrapping up the synapse and maintain the microenvironment
3: taking up K^+

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SAQ 2

Describe the changes that may occur when microglia are activated. [5 marks]

1: change in morphology

2: proliferation

3: TNF & IL-1 beta secretion

4: NO synthase generation

5: neuroinflammatory reaction

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